

Carlos Georges

California | carlos.georges@columbia.edu | +1 (702) 824-2468
github.com/georgc4

Summary

Linux Kernel Engineer at Intel delivering usable Linux platforms and the infrastructure that makes them reproducible for other engineers. Experience spans Ubuntu/Buildroot distributions, kernel packaging, graphics and boot enablement, pre-silicon environments, and hardware qualification. Combines end-to-end systems ownership with an accepted upstream ARM64 contribution and four years of production silicon and EDA experience.

Experience

Intel Corporation — Linux Kernel Engineer

June 2026–Present

Linux distribution and platform engineering

- Delivered Ubuntu Desktop and cloud platform support through containerized image assembly and Debian kernel packaging; established four release/debug variants with successful physical-board validation.
- Turned custom kernels and root filesystems into graphics-capable development environments by integrating Intel Xe firmware/drivers and Mesa/Weston/X11, with reliable boot services, desktop support, and A/B updates.
- Established Linux bring-up in local Simics and supported pre-silicon hybrid FPGA platforms through custom kernel configurations and repeatable firmware-to-kernel boot flows.
- Delivered self-contained Android kernel and OS build environments driven by exact source manifests, with successful builds for two target configurations; added profile customization and complete artifact delivery.
- Built a shared hardware-qualification and release system connecting platform images, exclusive board access, automated recovery, test evidence, and source provenance to traceable release candidates.
- Unified fragmented build and delivery tools across two repository organizations using Python, Bash, Docker, and GitHub Actions; provided self-service builds, results dashboards, and documentation for internal teams.

Intel Corporation — Physical Design & EDA Engineer

May 2022–June 2026

Production engineering tools and silicon methodology

- Modernized shared EDA workflows with modular, backward-compatible Tcl interfaces and production regression coverage, making implementation and validation logic reusable across existing flows.
- Built production QA and design-health reporting for timing, connectivity, and multi-voltage behavior; restored metrics-ingestion infrastructure during tapeout-critical outages.
- Owned GPU power-connectivity and UPF-aware implementation methodology; supported new-CPU RTL-to-GDS bring-up through shared Tcl/Perl flows and resolved power-grid, DRC, and IR-drop issues with cross-functional teams.

Selected Systems Projects

PS4 Linux Platform and Inference Infrastructure

Independent project — September 2026–Present

- Repurposed PS4 hardware into a custom Linux/Buildroot development environment, combining boot-loader, kernel, and AMDGPU changes with network access and A/B rootfs recovery.
- Created a source-pinned cross-build and deployment flow for x86-64 images from ARM64 hosts; packaged a CPU-targeted `llama.cpp` runtime and reproducible benchmark tooling for inference experimentation.

RV32 Linux SoC and Build Pipeline

Independent project — September 2026–Present

- Built an original Linux-capable RV32 SoC and SBI firmware, connecting hardware, device-tree, kernel, and BusyBox images into a complete boot flow that reached `/init` in Verilator simulation.
- Established reproducible image generation and automated regression coverage for processor, memory-translation, and peripheral behavior.

ARM64 Hypervisor Initialization Fix

Merged upstream contribution — AsahiLinux/m1n1#624, July 2026

- Contributed an accepted C fix to m1n1's ARM64 hypervisor, correcting secondary-CPU initialization ordering and synchronization for Apple M3 and later systems.

Simulation-Based CPU Power Optimization

Columbia University

- Integrated a reinforcement-learning model and an offline-trained lookup table into gem5 for selective functional-unit power gating; reported 10–20% energy savings under a custom simulation-based power model.

Education

Columbia University

May 2026

Master of Science, Electrical Engineering

University of Nevada, Las Vegas

December 2022

Bachelor of Science, Computer Engineering

Technical Skills

Linux and packaging: Ubuntu, Buildroot, BusyBox, Debian kernel packaging, kernel configuration, GRUB, A/B updates, systemd, device trees, graphics/firmware integration, boot and userspace debugging.

Languages: Python, Bash/Shell, C, Tcl, Perl, Verilog, RISC-V assembly, Make.

Build and delivery: Git, GitHub Actions, Docker, Git LFS, Artifactory, SFTP, manifest-driven builds, matrix workflows, artifact checksums and provenance, cross-compilation, release automation.

Validation and enablement: Hardware-in-the-loop testing, device leasing and recovery, regression testing, CodeQL, Semgrep, developer documentation, Simics, hybrid FPGA kernel configurations, FPGA/emulation build infrastructure, Verilator.

Hardware foundation: gem5, multi-voltage power domains, UPF, RTL-to-GDSII flows, IR-drop analysis, Synopsys and Cadence implementation tools.